

GREATER KANKAKEE, IL, USA

WMO: 722127

Lat: 41.072N

Lon: 87.847W

Elev: 629

StdP: 14.36

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04880

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.3	(c) 3.0	(d) -9.6	(e) 3.4	(f) -0.8	(g) -4.8	(h) 4.4	(i) 4.5	(j) 27.3	(k) 29.6	(l) 25.6	(m) 27.6	(n) 6.1	(o) 280	(p) 0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	92.5	75.1	90.2	73.8	87.8	72.9	78.5	88.0	77.0	86.0	75.3	83.6	10.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	136.5	84.4	73.9	129.6	82.2	72.6	123.7	80.5	42.7	88.1	40.7	86.5	39.2	83.4	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 24.9	(b) 21.3	(c) 18.9	(e) -10.9	(f) 95.4	(g) 7.1	(h) 3.2	(i) -16.0	(j) 97.7	(k) -20.2	(l) 99.6	(m) -24.2	(n) 101.4	(o) -29.4	(p) 103.7		
(4) 24.9	(5) 21.3	(6) 18.9	(7) -11.0	(8) 81.6	(9) 6.7	(10) 1.6	(11) -15.8	(12) 82.8	(13) -19.7	(14) 83.8	(15) -23.5	(16) 84.7	(17) -28.4	(18) 85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	24.3	27.5	39.8	50.9	62.1	71.1	74.4	72.4	66.8	54.1	41.3	30.4
	DBStd	19.85	11.82	12.13	11.15	9.13	8.93	6.21	6.03	5.86	7.48	9.39	9.59	10.97
	HDD50	2856	797	632	356	98	8	0	0	0	61	292	612	
	HDD65	5936	1261	1050	786	428	164	16	4	11	66	363	713	1074
	CDD50	3363	2	3	39	125	384	634	757	693	505	187	30	4
	CDD65	963	0	0	3	5	75	201	296	239	121	23	0	0
	CDH74	9385	0	0	29	99	820	1862	2990	2199	1147	238	1	0
	CDH80	3382	0	0	3	8	274	648	1196	783	402	68	0	0
Wind	WSAvg	8.0	9.6	9.3	9.5	10.4	8.5	7.0	5.5	4.8	5.7	7.9	9.0	8.9
Precipitation	PrecAvg	37.4	2.3	1.7	2.3	3.6	4.3	3.9	4.8	4.0	2.7	3.0	2.8	2.1
	PrecMax	51.1	5.0	4.9	4.3	7.2	7.0	9.6	11.0	9.4	7.9	7.2	6.0	4.3
	PrecMin	30.5	0.5	0.4	0.9	1.5	0.6	0.0	2.1	1.3	0.4	0.6	0.1	0.3
	PrecStd	5.5	1.3	1.2	1.1	1.5	1.9	2.2	2.5	2.3	2.0	1.7	1.5	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.5	62.9	78.8	81.3	91.1	93.4	97.2	93.4	93.0	87.9	71.4	61.4
		MCWB	57.1	55.5	63.9	62.8	70.8	73.4	79.1	75.4	72.4	71.2	58.8	57.3
	2%	DB	50.4	55.2	69.5	77.0	87.5	90.3	92.8	90.6	89.4	80.8	64.4	55.1
		MCWB	48.1	50.1	58.5	60.8	69.3	73.2	77.4	74.6	72.5	64.8	56.9	51.5
	5%	DB	44.8	47.8	63.2	72.7	82.4	87.4	90.1	88.0	84.3	74.9	61.0	50.0
		MCWB	42.0	44.3	55.3	58.7	66.2	72.4	75.4	73.5	70.2	63.5	54.4	47.4
	10%	DB	39.2	42.8	57.1	67.9	78.7	83.5	87.0	84.1	81.1	70.1	55.9	45.2
		MCWB	37.0	39.3	50.9	56.6	65.1	70.3	73.9	72.2	67.9	61.9	50.3	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.9	58.3	64.5	66.8	74.3	78.4	81.9	80.1	77.3	73.1	62.3	58.7
		MCDB	59.9	61.1	75.5	75.5	85.5	87.6	92.5	89.1	89.2	83.5	66.8	60.8
	2%	WB	48.6	51.0	60.5	63.5	71.4	76.5	79.5	77.8	74.7	69.5	58.9	52.8
		MCDB	49.7	54.9	67.8	72.4	82.1	85.4	89.5	86.6	84.4	76.7	63.8	54.7
	5%	WB	41.9	44.6	56.8	60.5	69.2	74.5	77.7	76.0	72.7	66.4	55.1	47.2
		MCDB	45.0	47.3	61.8	69.6	78.5	83.0	86.8	83.9	80.6	70.9	59.8	49.1
	10%	WB	37.2	39.1	51.4	58.2	67.2	72.7	75.9	74.3	70.7	62.5	51.3	42.4
		MCDB	39.1	41.8	57.0	67.0	75.7	80.9	84.2	81.6	77.7	68.8	56.0	44.7
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.7	18.2	21.6	21.8	20.4	20.4	20.9	22.4	20.2	16.9	13.3
		MCDBR	18.7	22.3	25.3	28.7	26.8	23.7	22.6	23.6	25.6	26.3	23.1	18.1
		MCWBR	16.5	18.0	16.8	16.4	12.6	10.5	10.3	10.5	10.8	13.3	16.0	16.0
	5% WB	MCDBR	18.4	21.5	22.7	25.3	22.5	20.0	20.4	19.3	21.0	20.6	19.9	16.9
		MCWBR	16.7	18.5	16.3	16.4	12.2	10.6	10.7	10.2	10.4	12.7	15.5	15.8
Clear-Sky Solar Irradiance	taub	0.296	0.323	0.352	0.405	0.440	0.441	0.460	0.452	0.421	0.361	0.318	0.296	
	taud	2.426	2.344	2.359	2.221	2.177	2.218	2.157	2.168	2.227	2.397	2.469	2.450	
	Ebn at Noon	273	280	284	275	267	266	259	258	257	263	263	264	
	Edh at Noon	25	32	35	43	46	45	47	45	40	30	24	23	
All-Sky Solar Radiation	RadAvg	540	820	1147	1482	1690	1937	1935	1702	1426	954	622	452	
	RadStd	56	96	85	119	143	125	98	97	115	100	60	51	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page